

Extended Abstract

Motivation Current literature prioritizes RGB inputs as semantically sufficient for goal-conditioned navigation, acknowledges that depth can potentially be a useful feature without insight on which specific environments and scenarios depth is useful, and reveals that depth data through LiDAR is computationally expensive for geometric inference. While recent visual foundation models suggest that RGB observations alone may provide sufficient information for policy learning, classical navigation systems such as SLAM have relied heavily on explicit geometric representations derived from depth sensing and recent works suggest depth as a potentially useful feature. To this end, we investigate whether depth provides a meaningful advantage across IL, offline RL, online RL, and model-based RL settings when combined with RGB observations in goal-conditioned navigation.

Method We study the impact of observation modalities on various RL algorithms by evaluating four main input modalities: RGB, RGB+Depth, RGB+BlackDepth, and Depth Only. We build upon LiMo, a goal-conditioned trajectory generation framework for ANYmal D (with GrandTour dataset). First, we establish an imitation learning baseline using behavior cloning. Second, we perform offline policy improvement using an Advantage-Weighted Regression (AWR)-inspired approach, where we sample candidate trajectories, evaluate their rewards, and update policy to higher-value trajectory chunks. Third, we evaluate online RL using PPO in Habitat’s PointNav, enabling policies to learn through environment interaction. Finally, we implement a MOPO-inspired model-based framework that learns latent dynamics and performs uncertainty-aware policy optimization.

Implementation For the baseline, we train our policies using frozen visual encoders and a transformer-based trajectory decoder that predicts 50 SE(2) waypoint sequences conditioned on the RGB goal image and the goal pose. For offline RL, AWR generates trajectory perturbations using either a standard navigation reward or a geometry-aware traversability reward. PPO policies are trained in Habitat PointNav using dense progress rewards and terminal success bonuses. Our MOPO-inspired framework learns the latent transition dynamics, reward models, and uncertainty estimates, which enables pessimistic value estimation over model-generated rollouts.

Results Across the offline and online RL settings, RGB+Depth consistently produces stronger policies than RGB-only observations. In imitation learning and AWR, we observe that RGB+Depth achieved lower validation loss and faster generalization and convergence. For AWR, the geometry-aware reward shaping objective also amplified the advantage of depth as an input feature by approximately 3.6x. With PPO in Habitat, RGB+Depth achieved the highest success rates and SPL, revealing promise to outperform SOTA with more diverse experimentation (PointNav benchmark). In contrast, for model-based RL, MOPO experiments revealed that Depth-Only observations significantly improved the actor (174 vs 49) and critic (12.40 vs 2.13) optimization, which resulted in lower training losses. We also outperform the IL validation loss baseline with 3x fewer training steps.

Discussion First, on average, depth improves state representations for policy learning, leading to stronger navigation performance and improved generalization. Second, geometry-aware rewards increase the feature value of depth by explicitly linking the observations to traversability, revealing a unique understanding of how representational learning can influence reward design. Third, for MOPO, our results reveal that geometric observation can simplify the value-function learning process and model-based optimization through more compact representation of geometric information. Nonetheless, improved optimization doesn’t always translate to optimal navigation; this results in an interesting tradeoff between representational richness and success in goal-conditioned navigation. We address this by still maintaining both RGB and Depth as input features.

Conclusion We present a comprehensive study spanning imitation learning, offline RL, online RL, and model-based RL with MOPO, revealing how including depth in our observational modalities can improve the policy performance (combined with RGB observations) and that geometry-aware rewards can also significantly increase the benefits of extracting depth signals. Furthermore, depth-only representations reveal signs of promise with more efficient value learning and policy optimization in model-based RL. Overall, our findings demonstrate that depth can serve as a meaningful input feature in various environments and can improve the effectiveness of navigation policies in benefiting from both semantic and geometric information.

Depth-Informed Visual Navigation Policies for Goal-Conditioned Trajectory Generation

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Abstract

Visual navigation policies have recently benefited from large-scale visual representational learning with RGB images, yet struggle with diverse edge-case scenarios. This motivates the question of whether RGB observations alone are sufficient for extracting relevant features in goal-conditioned embodied navigation. With classical robot navigation systems such as SLAM relying on geometric representations derived from depth sensing, previous embodied goal-conditioned navigation literature emphasizing the RGB modality as sufficient, and other depth sources such as LiDAR being computationally expensive to collect/infer, we investigate the role of depth across a spectrum of policy methods. We implement imitation learning as a baseline, AWR-inspired approach with two varied reward functions to extract meaningful depth, PPO in Habitat, and a variant of model-based offline policy optimization (MOPO). We build upon LiMO, a goal-conditioned trajectory prediction framework for ANYmal D, evaluating four observation modalities: RGB, RGB+Depth, Depth-Only, and RGB+BlackDepth. Across imitation learning, offline RL, and online RL, RGB+Depth consistently improves navigation performance relative to RGB-only policies, achieving lower validation losses, higher success rates, improved SPL, and lower final goal distances. Furthermore, reward functions that explicitly encode geometric traversability amplify the benefit of depth, increasing the performance gap between RGB and RGB+Depth by 3.6x. In contrast, for MOPO, Depth-Only observations often simplify optimization and yield lower actor and critic losses in model-based RL. Our results demonstrate that depth is more effective when combined with semantic RGB representations and when policy objectives require geometric reasoning, especially for edge-case scenarios. We outperform LiMo’s baseline performance with a lower validation loss at 3x less training steps. Additionally, our SPL and success rates in Habitat’s online performance suggest SOTA potential with drastically higher SPL metrics with RGB+Depth, when compared to PointNav’s benchmark. These findings challenge existing works that view modern visual foundation models as sufficient without depth and suggest the potential of geometry-aware multimodal policies for robust embodied navigation in diverse environments.

1 Introduction

Reinforcement learning (RL) has emerged as a powerful framework for training embodied agents capable of making long-horizon decisions from high-dimensional sensory observations. Recently, advances in offline reinforcement learning, online policy optimization, and model-based reinforcement learning have enabled increasingly capable navigation systems that learn directly from observations and actions. Specifically, within RL for embodied agents, visual navigation has emerged as a central problem, requiring robotic agents to reason about spatial goals in diverse environments such as outdoor and indoor environments. Recent advances in foundation vision models have improved visual

representational learning, enabling visual navigation policies to leverage rich semantic information from RGB imagery alone. For example, in Less Is More (Inglin et al.), the authors leverage classical geometric planners to generate synthetic trajectories for a transformer-based visual navigation policy that, conditioned on a single RGB image and goal pose, predicts goal-conditioned SE(2) trajectories and performs successfully in a variety of environments.

Classical navigation systems such as SLAM (Cadena et al. (2016)) have long relied on explicit geometric representations, which motivates the widespread use of depth sensors and RGB-D perception. In contrast, recent advances in visual foundation models have proposed a competing hypothesis: sufficiently powerful RGB representations may already capture much of the geometry that is required for successful navigation (Banerjee et al. (2022)). Despite this competing view, current literature has not studied 1) how 2D RGB and depth images can be concatenated as input features for goal-conditioned navigation with ANYmal D, 2) how RGB and depth as input features impact goal success and success weighted by inverse path length (SPL) across frameworks such as IL, AWR, PPO, and MOPO (i.e., IL as a baseline and from offline to online RL’s spectrum) and 3) whether depth can serve as a meaningful feature for stability and generalization beyond the fact that it introduces increased architectural capacity through more parameters. Most notably, to this end, existing work does not conduct rigorous ablations for 2D RGB+Depth comparisons in multiple RL frameworks.

In this work, we study the role of sensory modalities across the imitation learning (baseline) to RL spectrum. We build upon LiMO, a goal-conditioned navigation policy that predicts future SE(2) trajectories from an RGB goal image and goal pose and evaluate performance across four main modalities: RGB, RGB+Depth, RGB+BlackDepth, and Depth only. The use of BlackDepth preserves the complete multimodal architecture while replacing depth observations with a constant signal, which enables us to understand the contribution of depth relative to the additional modal capacity (i.e., black depth doesn’t contribute any meaningful features, but increases model capacity).

We then evaluate these modalities under four distinct policy-learning paradigms. First, we study imitation learning on a large-scale real-world navigation dataset GrandTour. Second, we perform offline policy improvement using an IQL-inspired objective that perturbs trajectory predictions, ranks them, and filters candidate trajectories through a reward function. Third, we evaluate online reinforcement learning through IL and PPO in Habitat, a high-performance 3D simulator, where policies directly influence future observations through online environment interaction. Finally, we investigate model-based offline RL using MOPO, where we combine latent dynamics models with policy optimization.

Our results suggest that across imitation learning, offline RL, and online RL, concatenating RGB+Depth features consistently tends to achieve the strongest navigation performance. However, in model-based RL, depth only representations substantially simplify the actor and critic optimization with significantly reduced loss. Furthermore, conducting geometry-aware reward shaping amplifies the benefit of depth by approx. 3.6x, revealing that the usefulness of geometric perception is determined not only by the observations that are available to the actual agent (i.e., input modalities), but also by how those observations can be exploited during optimization to obtain relevant and meaningful signals. These findings suggest a shift from asking whether depth is simply useful or not to actually understanding in which applications and why depth can serve as a useful feature, both quantitatively and qualitatively as we later outline.

2 Related Work

Goal-conditioned visual navigation has recently benefited from advances in visual representation learning and foundation models. For example, LiMo Inglin et al. (2026) demonstrates that transformer-based policies conditioned on a goal image and pose can generate navigation trajectories with limited real-world data episodes. This suggests that strong RGB representations may be sufficient for scalable navigation. Furthermore, more broadly, foundation vision models such as DINOv2 have enabled robust semantic feature extraction from RGB observations Oquab et al. (2024). In contrast, classical navigation systems, such as SLAM-based approaches, rely heavily on explicit geometric reasoning from depth and 3D scenes. It is also computationally expensive to infer depth from LiDAR inputs. This poses the open question of whether 2D depth images provide additional value beyond the modern RGB observations in current goal-conditioned navigation frameworks. Additionally, literature (Banerjee et al. (2022)) has revealed RGB image inputs (e.g., semantics) and ego views to

be impressive for object goal navigation (i.e., outperforming humans) and RGB to work extensively well for target-driven visual navigation (Zhu et al. (2016)).

Embodied navigation research has commonly established the success weighted by the inverse path length (SPL) as a metric for observing path success/efficiency Anderson et al. (2018). Our work builds upon imitation learning, offline, online, and model-based reinforcement learning. We leverage Advantage-Weighted Regression (AWR) Peng et al. (2019), which performs policy improvement through advantage-weighted behavior cloning, and Proximal Policy Optimization (PPO) Schulman et al. (2017), a widely adopted on-policy optimization algorithm. We also draw inspiration from MOPO Yu et al. (2020), which combines learned dynamics models with uncertainty-aware policy optimization. Unlike prior literature, we systematically evaluate RGB and depth modalities across all four learning paradigms, providing a clear insight into when geometric information improves navigation performance and the policy optimization process. We introduce novelty in this project by 1) leveraging 2D depth images as a relevant input image features for goal-conditioned navigation, 2) exploring the capabilities for depth to improve policy learning and generalization across various frameworks such as IL, AWR, PPO, and MOPO, and 3) how designing rewards with specific RL frameworks such as AWR can allow for more meaningful and geometrically rich information to be extracted in policy learning.

3 Method

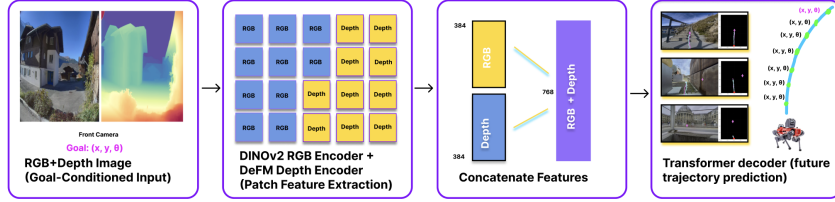


Figure 1: Method Overview for Imitation Learning Baseline

We implement Imitation Learning as a baseline, AWR with an IQL-inspired objective with two distinct rewards (one that explicitly prioritizes depth), IL and PPO in Habitat (PointNav), and a MOPO-inspired algorithm that allows us to penalize uncertain trajectory predictions.

Imitation Learning (Baseline): We trained baseline IL policies on GrandTour using supervised behavior cloning. Each sample consists of a front-camera observation, an SE(2) goal in a robot-local frame, and a 50-step expert SE(2) path. The policy encodes RGB observations with a frozen DINOv2 ViT-S/14 backbone for RGB images and for depth images, it leverages Depth-Anything and DeFM with a frozen DeFM ViT-S/14 backbone. The patch tokens are augmented with learned spatial embeddings and decoded by a 4-layer transformer using goal-conditioned time queries to predict the 50 waypoints.

We define the policy as

$$\hat{s}_{1:N} = \pi_{\theta}(I, g), \quad \hat{s}_t = (\hat{x}_t, \hat{y}_t, \hat{\theta}_t).$$

We formulate the training dataset as

$$\mathcal{D}_{\text{aug}} = \mathcal{D}_{\text{TEL}} \cup \mathcal{D}_{\text{GEO}}.$$

The behavior cloning objective is

$$\mathcal{L}_{\text{BC}}(\theta) = \mathbb{E}_{(I, g, s_{1:N}) \sim \mathcal{D}_{\text{aug}}} \left[\frac{1}{N} \sum_{t=1}^N \|\pi_{\theta}(I, g)_t - s_t\|_2^2 \right].$$

For waypoint regression, we use the mean squared error

$$\mathcal{L}_{\text{BC}}(\theta) = \frac{1}{N} \sum_{t=1}^N \|\hat{s}_t - s_t\|_2^2.$$

Advantage Weighted Regression (AWR)-Inspired Approach: We fine-tune the imitation policy with an AWR reward-filtered behavior cloning objective over offline trajectory candidates. For each of the samples, the policy first predicts a 50-step SE(2) trajectory, then generates 8 Gaussian-perturbed variants around this prediction and appends the expert trajectory. Each candidate was scored with a navigation reward that prioritized either 1) goal-progress, smoothness, and success or 2) the same reward, extended with ground-truth elevation geometry penalty for drastic drop-offs, rough terrain, or unknown cell locations. A value head was trained with 0.7 expectile regression against the set of candidate rewards, biasing towards high-reward trajectories. The policy was trained by MSE regression toward the single highest-reward candidate. The resulting loss was the filtered BC loss plus a weighted value loss.

$$R_{\text{goal}}(\tau, g) = -\|\tau_T^{xy} - g^{xy}\|_2. \quad (1)$$

$$R_{\text{smooth}}(\tau) = -\sum_{t=2}^T \|\tau_t - \tau_{t-1}\|_2^2. \quad (2)$$

$$R_{\text{base}}(\tau, g) = w_{\text{goal}} R_{\text{goal}}(\tau, g) + w_{\text{smooth}} R_{\text{smooth}}(\tau) + B_{\text{success}} \mathbf{1} [\|\tau_T^{xy} - g^{xy}\|_2 < d_{\text{success}}]. \quad (3)$$

$$R_{\text{geom}}(\tau, E) = -\sum_{t=1}^T \phi(E(\tau_t)), \quad (4)$$

$$R_{\text{geom_total}}(\tau, g, E) = R_{\text{base}}(\tau, g) + w_{\text{geom}} R_{\text{geom}}(\tau, E). \quad (5)$$

Proximal Policy Optimization (PPO) in Habitat: We evaluate RGB, RGB+Depth, RGB+BlackDepth, and Depth only in Habitat PointNav as a controlled simulation analogue of offline experiments with GrandTour. Each of these policies received an egocentric observation and a goal and selected one of the discrete navigation actions (e.g., move_forward, turn_left, turn_right, stop, etc.). We used a LiMO-style goal-conditioned policy that was adapted to PointNav. Visual observations were encoded with a frozen ResNet18 backbone, which produced spatial feature tokens that were then augmented with learned positional embeddings and passed through a Transformer decoder. We implement PPO in Habitat’s interactive PointNav environment. At every step, the policy samples an action from

$$\pi_{\theta}(a_t \mid o_t, g_t)$$

Then, the simulator transitions to the next viable state, and the agent receives a reward. The reward specifically encouraged progress toward the goal, including a small step penalty, and provided a sparse terminal success bonus for arriving near the goal at the end of the episode (penalizing false STOP actions taken by the agent). Thus, we define the reward as:

$$r_t = d(s_t, g) - d(s_{t+1}, g) - 0.01 + B_{\text{success}} \mathbf{1}[\text{success}_t].$$

We also optimize the clipped surrogate objective with a learned value baseline:

$$L_{\text{PPO}} = -\mathbb{E}_t \left[\min \left(\rho_t A_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) A_t \right) \right] + c_v \mathbb{E}_t \left[(V_{\theta}(s_t) - R_t)^2 \right] - c_e \mathbb{E}_t \left[H(\pi_{\theta}(\cdot \mid s_t)) \right].$$

where:

$$\rho_t = \frac{\pi_{\theta}(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$$

A_t is estimated from the rollout returns using the critic.

Model-based Offline Policy Optimization-Inspired Algorithm: We implemented a LiMO-style MOPO through a model-based offline RL method that uses actor-critic updates and learned latent dynamics ensemble. We convert each dataset sample into an offline translation:

$$(s_t, a_t, r_t, s_{t+1}, \text{done}_t)$$

Where s_t , is the LiMO visual-goal latent embedding, a_t is the expert SE(2) trajectory chunk that is converted into body-frame controls, and r_t is a navigation reward from goal progress, path length, smoothness, and success. The learned ensemble predicts:

$$\Delta s_t, \hat{r}_t, \widehat{\text{done}}_t = f_m(s_t, a_t), \quad s_{t+1} = s_t + \Delta s_t.$$

The policy in this algorithm is a Gaussian actor over full 50-waypoint control sequences. We initialize with the IL policy by using the LiMO trajectory prediction as a prior and the actor learns the residual:

$$a_t = a_{\text{prior}}(s_t) + \delta_\theta(s_t)$$

The critics are trained with SAC targets on real offline transitions and model-generated synthetic transitions. The synthetic rewards are pessimistic:

$$U(s, a) = \frac{1}{M} \sum_{m=1}^M \|f_m(s, a) - \bar{f}(s, a)\|_2^2, \quad \bar{f}(s, a) = \frac{1}{M} \sum_{m=1}^M f_m(s, a).$$

$$r_{\text{pess}}(s, a) = r_{\text{model}}(s, a) - \lambda U(s, a).$$

The actor serves to maximize the critic value with entropy regularization and a small BC prior penalty:

$$\mathcal{L}_{\text{actor}} = \mathbb{E}_{(s,a) \sim \mathcal{D}} [\alpha \log \pi_\theta(a | s) - Q_\phi(s, a)] + \lambda_{\text{BC}} \|\mu_\theta(s) - a_{\text{prior}}(s)\|_2^2$$

In comparison to MOPO, our implementation is adapted towards LiMO, where the state is a learned LiMO visual-goal latent embedding, the action is a full 50-step trajectory, the actor is residualized around an IL trajectory prior, rewards are trajectory rewards from GrandTour data, the learned model predicts latent next-state reward and done (true/false), and we also add stabilizers that are not in the vanilla MOPO version such as CQL critic term and a BC anchor when unfreezing LiMO’s head.

4 Experimental Setup

We experimented with four main modalities: RGB, RGB+Depth, RGB+BlackDepth, and Depth Only, where RGB+BlackDepth serves as a control ablation to understand if performance gains can be attributed to meaningful depth features or increased architectural capacity (e.g., more parameters).

For the IL baseline, AWR, and MOPO, we train using LiMO’s offline GrandTour dataset with the four modalities. In AWR, we try nearby paths and imitate the best. In MOPO, we learn a world model and a critic, and optimize the policy against the predicted long-term value to penalize uncertain trajectories. We leverage 25+ missions from GrandTour, focusing on diverse outdoor and indoor environments.

For PPO in Habitat, we train using Habitat’s (Habitat-Matteport 3D Dataset) H3MD dataset on 20+ scenes.

For metrics that enable us to infer if depth serves as a meaningful input feature, we study the success of the trajectory, the success weighted by inverse path length (SPL), the validation loss, MAE, and the qualitative rollouts of trajectory predictions (e.g., smoothness comparisons based on yaw changes).

We define success as

$$\text{Success} = \frac{1}{K} \sum_{i=1}^K \mathbf{1} [d_i^{\text{final}} < d_{\text{success}}].$$

We define SPL as

$$\text{SPL} = \frac{1}{K} \sum_{i=1}^K S_i \frac{\ell_i}{\max(p_i, \ell_i)},$$

where S_i is the success indicator, ℓ_i is the shortest-path distance, and p_i is the executed path length.

We define final goal distance as

$$\text{FGD} = \frac{1}{K} \sum_{i=1}^K d_i^{\text{final}}.$$

We define MAE as

$$\text{MAE} = \frac{1}{KN} \sum_{i=1}^K \sum_{t=1}^N \|\hat{s}_{i,t} - s_{i,t}\|_1.$$

5 Results

We implement and train with imitation learning as a baseline, AWR with two modified reward functions (one that does not leverage depth and one that does), and MOPO. As we delineate in the 5.1, while depth only showed promising signs for validation metric and best metric, we observed greater goal distance, success, and SPL feasibility with RGB+Depth and given our motivation to build upon LiMo, we build upon proven results that reveal RGB as still meaningful, but depth as an additional, helpful feature.

5.1 Quantitative Evaluation

Table 1: Comparison of RGB and Depth-Based Navigation Models with (IL, AWR, AWR+Depth, MOPO)

Model	Val Metric @ 7k / Final	Best Metric	Steps to Best
RGB Only (IL)	0.259	0.255	~27k
RGB + Real Depth (IL)	0.243	0.243	~7k
RGB + Black Depth (IL)	0.255	0.246	~2k
Depth Only (IL)	0.246	0.235	~2k
RGB Only (AWR)	0.296	0.295	~4.5k
RGB + Depth (AWR)	0.286	0.285	~6k
RGB Only (AWR + Depth)	0.405	0.405	~19k
RGB + Depth (AWR + Depth)	0.451	0.355	~31k
RGB Only (MOPO)	3.03	2.96	~1k
RGB + Real Depth (MOPO)	3.13	3.09	~3k

We observe that for IL as a baseline, RGB+Real Depth performs significantly better than just RGB only with a validation metric of 0.243 compared to 0.259 (i.e., 6.18% less) and only requiring 7k steps, instead of 27k steps, to reach the metric. As RGB+Depth presents a significant improvement in generalization and learning depth features compared to the ablation control RGB+BlackDepth (i.e., 0.243 vs 0.255), we emphasize findings from RGB only vs RGB+Depth for AWR and MOPO.

Furthermore, for AWR, we experiment with two different reward functions. The first (1) reward function prioritizes goal-progress, smoothness, and success. The second (2) reward function prioritizes the same factors and is augmented with a ground-truth elevation geometry penalty (using traversability maps) for drastic drop-offs and rough terrain, to name a few. For reward (1), we observe that RGB+Depth enables a similar trend to IL with a lower validation metric of 0.286 compared to RGB Only’s 0.296 and only a 1.5k step difference. Furthermore, when we make depth more explicit with reward (2), we observe that RGB+Depth achieves a best metric of 0.355 compared to 0.405 (=12.35% difference). While RGB+Depth with reward (2) requires an additional 12k steps to reach a better validation loss metric, these results reveal promising signs that RGB+Depth, when appropriately shaped by a reward function that elicits depth’s helpful features in edge cases (e.g., facing glass),

can improve the policy’s ability to generalize and learn more semantically and geometrically rich information.

Below, we observe the performance of RGB, RGB+Depth, RGB+BlackDepth, and Depth on imitation learning as a baseline. While Depth only in Table 1 results in the best metric of 0.235 within 7k steps, we observe that RGB+D consistently outperforms depth only in terms of the goal distance (-4.4% difference), success (3.1% more successful), and SPL (7.7% more SPL).

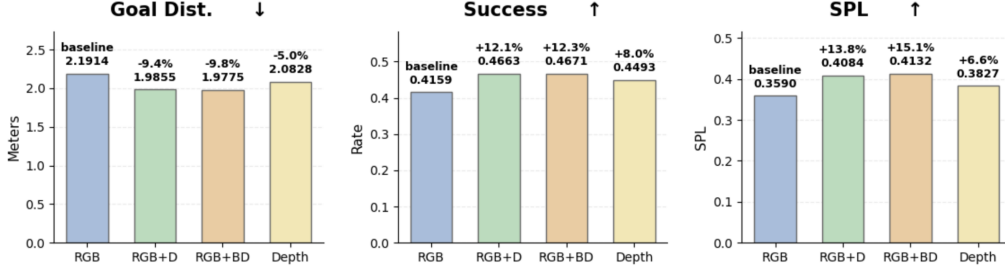


Figure 2: Comparison of Goal Distance, Success, and SPL across

For AWR with the 1st reward (i.e., goal smoothness, success, etc.), we observe below that RGB+Depth results in 3.5% less validation loss and a 4.9% less best validation MAE, revealing that RGB+Depth enables the model to potentially more strongly generalize and earlier on in training.

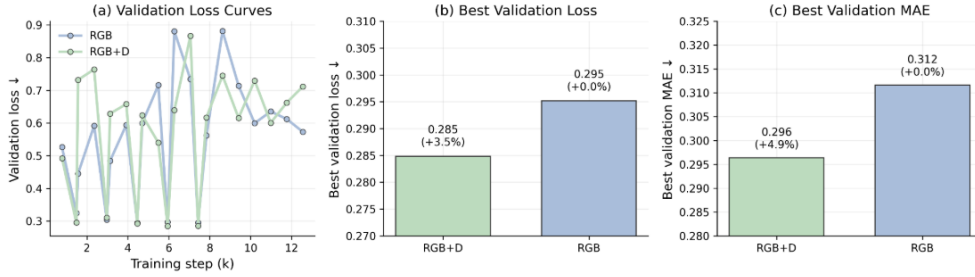


Figure 3: Comparison of validation loss curves, best validation loss, and best validation MAE with AWR with reward 1

For AWR with the 2nd reward (extracts depth features through traversability maps), we observe that RGB+D performs significantly better in terms of the best validation loss (12.3% difference compared to RGB only) and best validation MAE (5.7% difference compared to RGB only). Although the validation loss curve for RGB+D is initially unstable, we observe with more training steps, that it approaches a lower validation loss than all other modalities. This reveals that with more novel reward shaping approaches, we can explicitly extract relevant features from depth inputs to improve model generalization.

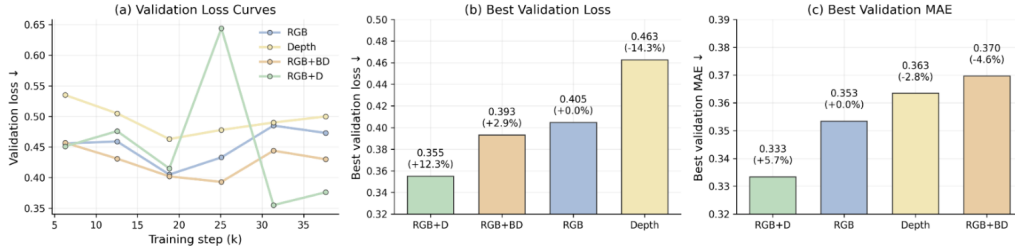


Figure 4: Comparison of validation loss curves,

In comparing the AWR with reward 1 to AWR with reward 2, we observe a significant increase in the benefit of depth (i.e., observing best validation loss of 0.405 for RGB and 0.355 for RGB+Depth).

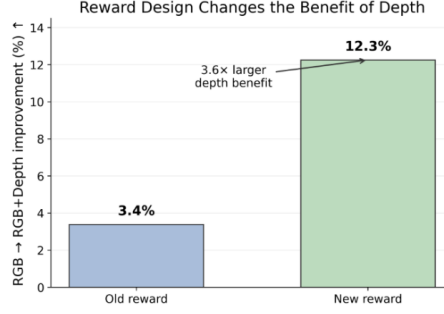


Figure 5: Depth benefit with AWR reward 2

Furthermore, in order to evaluate online RL under controlled, interactive conditions to understand if the RGB vs RGB+Depth comparison still holds, we deploy our IL pretrained checkpoint in Habitat with PPO. We observe that, on average, RGB+Depth enables the highest success rate of 0.9883, highest SPL of 0.9497, and lowest final distance to goal (i.e., if the agent is within 2 meters of the final goal) with 0.0930.

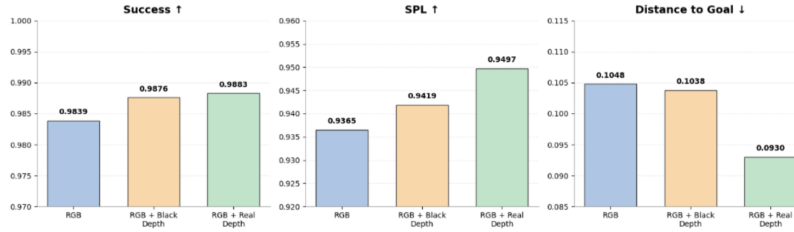


Figure 6: PPO in Habitat Metrics: Success, SPL, Distance to Goal

It is important to note that RGB+Depth does not guarantee greater success in all scenarios. We observe, as we later reveal in the Qualitative Analysis, that depth appears to be more helpful in edge cases where RGB can't geometrically reason about a scenario (e.g. a glass frame). In a few episodes below, we observe that RGB+Depth can enable greater success and SPL, but when the agent starts facing a direct occlusion, it tends to struggle and exploration is minimized (in comparison to RGB). We observe in episode 1, for example, that the agent isn't able to move due to starting in front of an obstacle. Nonetheless, RGB successfully reaches the goal through greater exploration (light gray).

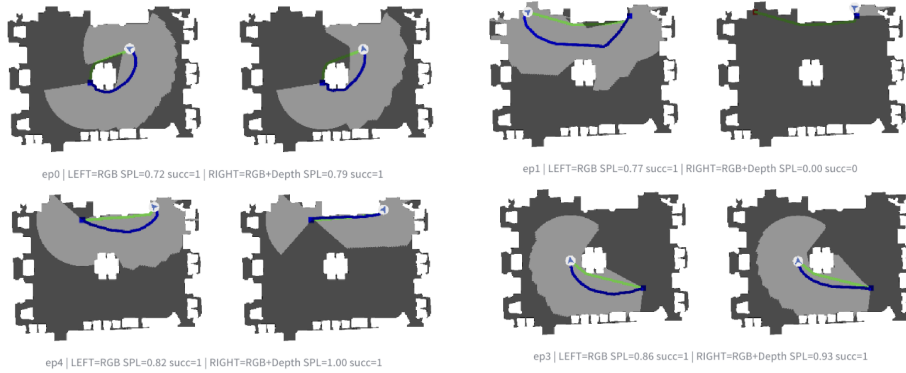


Figure 7: Habitat episode trajectories

For MOPO, we also observe the results from the optimization across the four modalities of RGB, RGB+Depth, RGB+BlackDepth, and Depth Only. We observe that depth-only is much easier for MOPO to optimize, achieving a drastically lower actor loss (48 vs 174-423) and critic loss (2.13 vs 12.4-19.5) than all of the RGB-based variants. This trend reveals that adding RGB, especially the control of RGB+BlackDepth, makes value-learning and policy-learning significantly harder. This is because depth provides a more compact representation of the geometry relevant for navigation and visual observations that are high-dimensional introduce more noise.

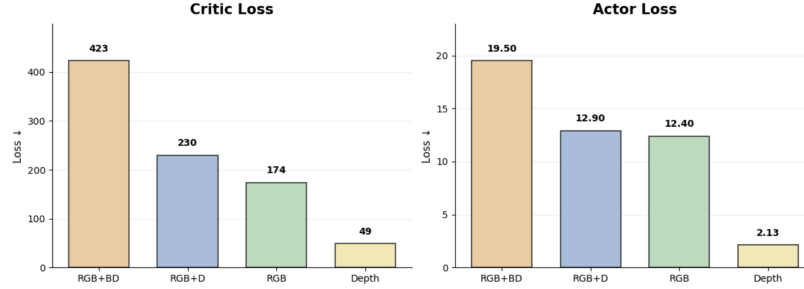


Figure 8: MOPO optimization comparison across modalities

5.2 Qualitative Analysis

Note that the qualitative analysis is not limited to this section and is also conducted in the above section with various graphs.

While the quantitative results establish that RGB+Depth generally improves navigation performance, a qualitative inspection of the predicted trajectories reveals when and why depth can be beneficial. We observe that depth as a feature primarily improves the behavior in geometrically ambiguous situations where RGB observations alone can't provide sufficient spatial cues. For example, these scenarios may entail transparent surfaces, narrow passages, abrupt terrain changes, and cluttered regions where the appearance doesn't reliably reveal traversability.

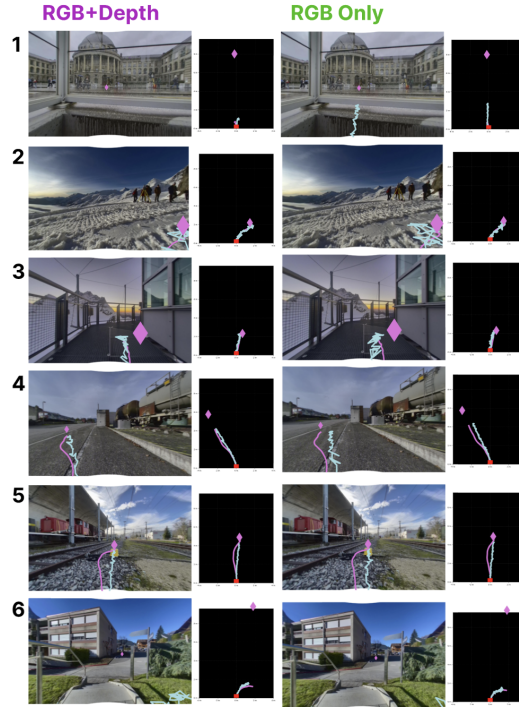


Figure 9: RGB vs RGB+Depth trajectory predictions

In these figures, the blue prediction represents the policy’s prediction and the pink is the ground-truth with the pink diamond representing the goal. For example, in Figure 9, we see that in the first example of a window being present, the RGB+Depth policy does not output a prediction that directly collides with the window, whereas the RGB only policy outputs a prediction that assumes it is safe to navigate through the window. This difference is stark and the use of depth can improve the safety of these four-legged agents. Furthermore, for the second example, we see that having depth as a feature allows for a more smooth trajectory that can navigate on the snow terrain (i.e., less frequent yaw changes and less scattered SE(2) trajectory waypoint predictions). For the third example, we see a similar performance where depth as an input feature helps the model output a more smooth trajectory. In example 3, similar smoothness is observed. In example 4, depth enables the prediction to arrive closer to the goal. In example 5, RGB+Depth enables a smoother trajectory with minimal yaw deviations compared to RGB only. In example 6, we observe RGB only performing better with a more smooth output, likely due to the traversability hindrance that depth may pose. Thus, this suggests that in environments where we are penalizing drastic changes in the traversability map, considering a goal in such a corresponding direction, depth may not generalize well or we may need to modify our reward function accordingly.

In reference to the figures outlined in 5.1, and to further elaborate on the qualitative analysis, for imitation learning and AWR, RGB-only trajectories frequently exhibit larger heading corrections and less direct paths toward the goal. In contrast, RGB+Depth predictions tend to produce more smooth trajectories with more consistent progress toward the target location. We see that this improvement is more pronounced when the reward function incorporates traversability information, which suggests that the policy is able to learn associate depth cues with terrain safety and navigation feasibility rather than merely exploiting the additional architectural capacity.

As shown in Figure 7, RGB+Depth often reaches the goal using shorter and more direct trajectories, which can result in improved SPL. However, depth as a feature does not universally improve performance as in several episodes, agents that are initialized near obstacles tend to exhibit reduced exploratory behavior. In these examples, RGB-only policies occasionally succeed by exploring other alternative routes in the environment that the depth-conditioned policies may not explore.

The MOPO experiments reveal a complementary trend in that depth-only representations produce substantially lower actor and critic losses than all RGB-based modalities, suggesting that geometric observations yield a simpler optimization modality. We anticipate further deploying these policies in four-legged robots in the future to further observe SPL, success, and other relevant metrics such as distance to final goal.

6 Discussion

Our experiments from offline to online RL reveal that depth can serve as a meaningful feature in embodied navigation. In contrast to the view that modern visual foundations can extract sufficient geometry from RGB images alone, we find that explicit depth information consistently improves the navigation performance when combined with RGB observations. Across imitation learning, offline RL, and online RL, RGB+Depth achieves more strong generalization, higher success rates, and more efficient navigation than RGB-only policies.

More importantly, the benefit of depth is not fixed and depends on the learning objective. We can’t anticipate depth to *always* be meaningfully helpful. In AWR, introducing the geometry-aware rewards had significantly increased the advantage of RGB+Depth over RGB, suggesting that redefining rewards to extract more useful geometric information must be leveraged in the optimization process instead of the raw input.

Furthermore, our results from MOPO reveal an interesting tradeoff: depth-only observations (or, on the spectrum, adding depth to our input), drastically simplifies the optimization, producing lower actor and critic losses than RGB-based modalities. As current literature reveals that optimization gains do not necessarily translate to the most optimal navigation performance, we hypothesize that geometry alone can provide an efficient representation for planning and value estimation based on our prior results and MOPO. RGB still plays a significant role in that it complements the depth input with semantic cues that are important for robust goal-conditioned navigation.

To address limitations, 1) our work focuses on building upon a primary navigation architecture with ANYmal D and it remains unclear how such performance will generalize to other navigation policies, foundation models, or even robotic platforms, 2) while we evaluate our results using offline and online learning paradigms, most of our experiments are conducted using fixed offline datasets or online simulations, which enables open questions for long-term deployment with long-horizon scenarios in diverse real-world environments, and 3) our depth observations are generated using monocular depth estimation, which can introduce estimation errors that can affect the measured benefits of geometric information.

For future work, the strong performance of RGB+Depth suggests that geometry and semantics can provide complementary information. Future work seeks to explore 1) adaptive fusion mechanisms to dynamically weigh RGB and depth in certain scenarios where RGB or depth is more helpful through an adaptive reward, 2) jointly designing observation modalities and reward functions instead of treating them independently based upon the observation that geometry-aware rewards can extract more depth, 3) exploring how natural language conditioning through VLAs can help ground the model in less uncertainty with edge-case scenarios in considering both input modalities, 4) exploring geometry-centric world models and latent representations that retain semantic information while also still preserving the simplicity of geometric planning, and 5) scaling our experiments to 100+ missions with extensively longer trainings with bigger compute budgets. All of our experiments were run using Modal’s H100s and at times, B200s to accelerate training.

7 Conclusion

In this project, we studied the role of depth across a spectrum of embodied navigation methods, ranging from imitation learning to offline and online reinforcement learning. We build upon LiMo, evaluating with four modalities, RGB, RGB+Depth, RGB+BlackDepth, and Depth-Only across IL, AWR, PPO, and MOPO-inspired frameworks. Our results demonstrate that explicit depth information consistently improves the navigation performance when it is combined with RGB observation, leading to stronger and earlier generalization, higher success rates, improved SPL, and more efficient learning/optimization. In our experiments, the RGB+BlackDepth control confirms that such difference gains are from meaningful geometric information instead of simply increased architectural capacity. Beyond these modality comparisons our findings reveal that the value of depth depends strongly on the learning objective. We observe that geometry-aware rewards substantially improve the benefits of leveraging depth as a feature. Furthermore, depth-only observations can simplify model optimization in model-based RL, highlighting a unique tradeoff between optimization efficiency and navigation performance. Overall, our results suggest that geometry and semantics play complementary roles in embodied navigation. Instead of replacing RGB representations, we encourage depth as a geometric prior that can improve reasoning in ambiguous or safety-critical (e.g., window) environments. We hope such findings motivate future work on geometry-aware multimodal policies, reward design, and models that can effectively integrate both semantic and geometric understanding for robust real-world goal-conditioned navigation.

8 Team Contributions

- **Group Member 1:** The sole author of this work conducted all experiments, data processing, and code implementations. Proposal, poster, milestone, and final report. I would like to acknowledge ASL for their guidance and compute resources.

Changes from Proposal We experiment beyond the imitation learning baseline with AWR, varied reward objectives, PPO in online environments (Habitat), and MOPO for model-based RL.

9 AI Tools Disclosure

Claude Code was sparsely leveraged solely for the purposes of debugging for training. We build upon prior works such as LiMO and all algorithms are implemented independently.

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A Additional Plots

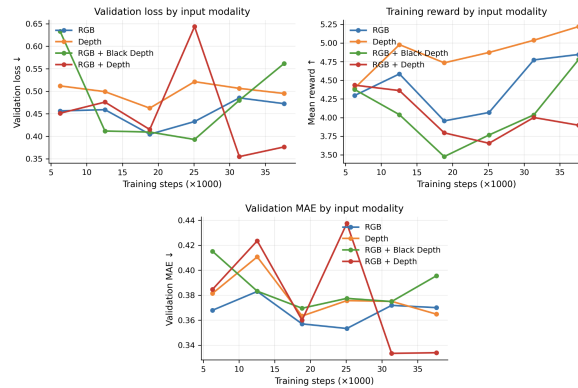


Figure 10: AWR-like approach with geometric reward

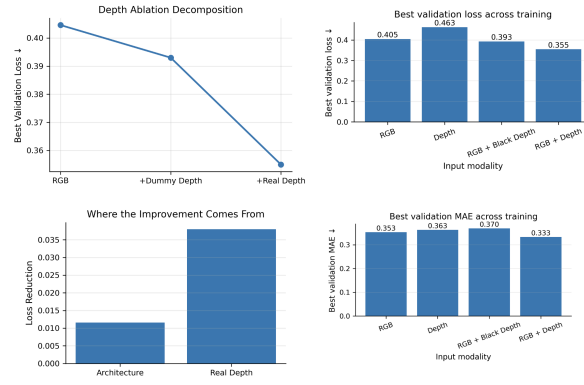


Figure 11: Bar charts comparison with AWR-like approach with geometric reward